

UNIT-5

Model Fitting in R

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Contents

- Regression Models
- Linear and Logistic Model
- Classification Models
- Decision Tree
- Naïve Bayes
- SVM
- Random Forest,
- Clustering Models – K Means and Hierarchical clustering

Regression Modelling

- Regression analysis is a statistical tool to estimate the relationship between two or more variables

Types of Regression in R:

There are mainly three types of Regression in R programming that is widely used. They are:

- Linear Regression
- Multiple Regression
- Logistic Regression
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Linear Regression

The Linear Regression model is one of the widely used among three of the regression types. In linear regression, the relationship is estimated between two variables i.e., one response variable and one predictor variable. Linear regression produces a straight line on the graph.

Mathematically

$$y = ax + b$$

Where,

x indicates predictor or independent variable

y indicates response or dependent variable

a and **b** are coefficients

Logistic Regression

Logistic Regression is another widely used regression analysis technique and predicts the value with a range. Moreover, it is used for predicting the values for categorical data. For example, Email is spam or non-spam, winner or loser, male or female, etc. Mathematically,

$$\frac{1}{1 + e^{-z}} \quad \frac{1}{1 + e^{-z}}$$

where,

y represents response variable

z represents equation of independent variables or features

Various Classifiers are:

- **Decision Trees**
- **Naive Bayes Classifiers**
- **K-NN Classifiers**
- **Support Vector Machines(SVM's)**

Popular clustering techniques

- K-means clustering
- Hierarchical clustering

References

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